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MILITARY SPECIFICATION

TRITIUM AIR MONITOR, PORTABLE OR FIXED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1 1 Scope This specification covers the design, production, performance requirements, and tests of portable or fixed air monitoring equipment, hereinafter referred to as the monitor, which is used for monitoring the level of tritium concentration in the air

2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6 2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests H I (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-W-4088	- Webbing, Textile, Woven Nylon.
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts) Packaging of

STANDARDS

MILITARY

MIL-STD-12	- Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	- Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification and Production.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-1399 Section 300	- Interface Standard for Shipboard Systems Section 300 Electric Power Alternating Current (Metric)

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MILITARY (Continued)

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-2036 - General Requirements for Electronic Equipment Specifications

HANDBOOKS

MILITARY

MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies.
MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2 1 2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
EE730-HA-OMI-010/8153 PDR73

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2 2 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3 1 First article When specified (see 6.2), a sample shall be subjected to first article inspection (see 6 4) in accordance with 4.4.

3 2 General The monitor shall be in accordance with MIL-STD-2036, except as stated in (a) through (e) and hereafter specified

(a) Quantitative value for hydrostatic pressure
(b) Inclination angle
(c) Airborne and structure noise limits.
(d) Time totalizing meter.
(e) Illumination

3 2 1 Solid-state circuitry Solid-state circuitry shall be employed unless otherwise approved by the contracting activity

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3.2.2 Parts and materials. Parts and materials shall be selected and applied in accordance with MIL-STD-2036, except plastic encapsulated microcircuits shall not be used

3.2.2.1 Microelectronic devices. Microelectronic devices used in the monitor design shall be in accordance with requirement 64 of MIL-STD-454.

3.2 2.2 Printed circuit boards (PCB's) PCB's within each module or assembly shall use plug-in construction to facilitate ease of removal and insertion

3 2.2 2 1 Module or PCB removal and insertion. The monitor shall be constructed so that removal or insertion of any module or PCB shall not be detrimental to the monitor

3 2 3 Compatibility with IC-T2/FC type converters. The monitor shall operate with the Navy type IC-T2/FC and IC-T2/FC(A) tritium converters to measure the tritium content in brine in accordance with NAVSEA EE730-MA-OMI-010/8153 PDR73

3 2 4 Connectors The monitor shall have one connector to provide for remote alarm and remote connections

3 2 5 Mounting The monitor shall be capable of being mounted on existing ship's mounting brackets

3 3 Construction requirements The construction requirements shall be as specified in 3 3 1 through 3 3 4

3 3 1 Equipment components The equipment shall consist of the following components

(a) One monitor
(b) One stowage case
(c) One power cable (three-wire, 10 meters (m) (32 8 feet (ft)) long
(d) Two remote sensing hoses
(e) One shoulder strap.
(f) Mounting assembly
(g) High pressure fitting
(h) Spare air filters
(i) Nickel cadmium (Ni-Cad) batteries

3 3 2 Stowage case A dripproof stowage case shall be provided with the monitor The capacity of the stowage case shall provide sufficient space for stowage of a power supply cord, remote sensing hose, shoulder strap, and technical manuals The stowage case shall be provided with the means for bulkhead mounting in accordance with MIL-STD-2036 The stowage case shall be made so that the equipment shall be removable from the front access panel The inside of the front panel shall be padded with at least 1-inch thick black ethafoam or equivalent

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3.3.3 Carrying strap. The carrying strap shall have closed-loop-type snap-on fasteners which will not detach from equipment unless manually released. The carrying strap shall be nylon in accordance with MIL-W-4088 and shall support a minimum of three times the weight of the equipment and shall be expandable to 1.37 ± 0.08 m (54 ± 3 inches (in)).

3 3 4 Remote sensing hoses. Two remote sensing hoses are required, one 0 914-m (3-ft) hose and one 6.056-m (20-ft) hose Hose material shall be typon tubing (or equal) 6.35 millimeters (mm) (0.25 in) in diameter.

3 4 Detailed requirements. The detailed requirements shall be as specified in 3 4.1 through 3 4.30

3 4 1 Monitor The monitor shall contain the air moving system, ion trap, detection chamber, signal converter, signal amplifier, audible alarm, readout meter, self-test circuit, electrical controls, and provision for operation from a power supply of 115 volts (V) plus or minus 10 percent alternating current (ac), 60 hertz (Hz) plus or minus 5 percent, single-phase The monitor shall be contained in a dripproof stowage case. Switching contacts shall be covered Relays shall be hermetically sealed.

3 4 2 Signal converter housing The precautions specified in (a) through (c) shall be provided for in the signal converter housing

(a) Signal converter housing shall be sealed at all points so as to withstand a minimum of 2 ± 0 2 pounds per square inch (lb/in^2) positive and negative pressure for a period of 15 minutes
(b) Electrical wiring associated with internal signal converter housing circuitry shall be self-contained within the housing
(c) Signal converter housing shall be provided with a readily replaceable desiccant capsule of a type and quantity of desiccant so as to provide protection to the high impedance circuitry for a minimum period of 1 year, with less than 50 percent saturation of the desiccator.

3 4 3 Signal converter The signal converter shall be as specified in 3 4.3 1 and 3 4.3.2

3 4 3 1 Electrometer tube If an electrometer is used, off-set current of the electrometer grid shall be 2 by 10^{-15} amperes (A), or less

3 4 3 2 Vibrating capacitor If a vibrating capacitor is used as a signal converter, the vibrating capacitor shall be given special attention in order to obtain maximum stability and efficiency Contact potential shall not exceed 90 millivolts (mV) Drift in contact potential shall not exceed 1.0 mV per day noncumulative. Insulator resistance shall be not less than 10^{15} ohms.

3 4 4 Power requirements The monitor shall operate from a self-contained Ni-Cad battery or 115 V, 60-Hz, single-phase, type I electric power in accordance with MIL-STD-1399, section 300. Battery operation may be obtained by means of a self-contained battery and selector switch or by automatic changeover from ac to battery in the event of ac power failure or removal Charging circuitry shall be contained either in the dripproof stowage case or as part of the equipment

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3.4.4 1 Power-on indication. Power-on indication shall be in accordance with MIL-STD-1472.

3 4.4.2 Power failure With the monitor operating on the most sensitive range and with the alarm point set at 80 percent of full scale or higher, interrupting the power supply and subsequently restoring it shall not cause a false alarm to sound upon restoration of power The normal background indication shall automatically resume in less than 2 minutes after power is restored

3 4 4 3 Fuses. The monitor shall be equipped with power supply fuses. Each side of the line shall be fused with indicating fuses. Fuse ratings shall be that standard rating which is equal to or greater than 125 percent of the monitor rated load.

3 4 4 4 Battery compartments Battery compartments shall contain rechargeable Ni-Cad batteries The battery compartment shall have vents to the outside of the monitor

3 4 4 5 Battery operation and charge rate The monitor shall operate on battery for a minimum of 4 hours without charge, with no degradation of readout accuracy The battery shall charge, after a 4-hour battery operation, to full charge within a maximum of 14 hours The battery charge circuit shall prevent the battery from charging at too high a rate or overcharging after the battery has received a full charge

3 4 5 Ranges The ranges shall be as specified in 3 4 5 1 and 3 4 5 2

3 4 5 1 Sensitivity ranges Four linear sensitivity ranges shall be provided as specified in (a) through (d) for monitoring tritium concentration in air Microcuries per cubic meter of air

(a) 0 to 10
(b) 0 to 100
(c) 0 to 1000
(d) 0 to 10,000

3 4 5.2 Display Tritium concentrations in microcuries per cubic meter shall be displayed on a suitable readout Readout shall be meter-type and shall be on a 63 5-mm (2 50-in) minimum diameter meter conforming to MIL-M-10304. Meter sensitivity shall be at least 50 microamperes full scale

3 4 6 Alarm An audible alarm shall be provided which will be actuated when the activity level is greater than a predetermined set point The alarm actuation level shall be capable of being set at any point above 25 percent of full scale on any range Once energized, the alarm shall continue to sound until the alarm reset switch is actuated by the operator. An alarm cutout switch shall also be provided Change in temperature shall not actuate the alarm

3.4.7 Background. The background radiation indication due to normal background gamma radiation and electrical noise shall not go below minus 5 percent (from zero mark) or above 20 percent (from zero mark) of full scale on the most sensitive range (0 to 10)

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3 4.8 Gamma compensation. Compensation shall be provided for readings which would be caused by uniform cobalt 60 or radium gamma ray fields up to 5 milliroentgens per hour In such a field, the meter indication shall not go below minus 5 percent (from zero mark) or above 5 percent (from zero mark) of full scale on the most sensitive range (0 to 10)

3 4.9 Accuracy. The normal operating accuracy for monitoring the tritium level in air under all conditions shall be within plus or minus 15 percent of full scale above background radiation Under controlled conditions, a calibration accuracy of plus or minus 10 percent of full scale above background radiation shall be obtained.

3 4 10 Response time The monitor shall indicate at least 63 percent of the actual tritium level within 1 minute after being exposed to a step function increase of tritium in air at a concentration of 80 to 100 percent of full scale on the most sensitive range

3 4 11 Warm-up time After 5 minutes warm-up, zero drift shall not exceed 50 percent of full scale during any 30-minute period In a 4-hour period, after 30 minutes warm-up, zero drift shall not exceed 30 percent of full scale on the most sensitive range (0 to 10)

3 4 12 Circuit check The monitor shall contain an electrical calibration check to verify complete end-to-end circuit operation This check shall provide a predetermined input to the extreme input end of the circuit which will provide a predetermined meter reading

3 4 13 Calibration control A calibration control shall provide an adjustment of plus or minus 25 percent of the normal sensitivity on all ranges Adjustment of the calibration control shall have no effect on the zero reading.

3 4 14 Air circulation system The air circulation system shall be equipped with a positive displacement type air pump so as to provide positive air circulation The air pump shall allow the use of a sensing hose (see 3.3.1) up to 6 096 m (20 ft) long, so that remote air monitoring of tritium concentration in air may be made The air circulation system shall be sealed so as to withstand a minimum of 1 lb/in^2 positive and negative pressure for a period of 15 minutes.

3 4 14 1 Air pump motor. The air pump motor shall be equipped with electronic noise filters to prevent spurious signals in the monitor, and shall utilize oil-less type, sealed bearings

3 4 14 2 Filters Filters shall be included to minimize build-up of foreign material in the ion trap Ease of filter change and access to spare filters shall be given primary consideration Filters shall filter particles larger than 2 micrometers (μm).

3 4.14 3 Air flow indicator. A panel-type indicator shall be provided to indicate that air flow through the monitor is above a minimum consistent with response time requirements (see 3.4.11). Considerations for air flow shall be addressed to such factors as filters, tritium air pulses, use of sniffer hose, and gross inaccuracies caused by extreme high rates of air flow. Metal shall be used to the maximum extent as structural support to preclude accidental breaking of the indicator

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3.4.15 Decontamination. The monitor shall be capable of self-decontamination to indicate less than 50 percent of full scale on the most sensitive range (0 to 10) within 30 minutes following a 15-minute exposure to contaminated air samples of 800 to 1000 microcuries of tritium per cubic meter

3 4 16 Electrostatic ion trap An electrostatic ion trap shall be provided between the filter and detecting element Field strength for the electrostatic ion trap shall be not less than 20,000 volts per meter (V/m)

3 4 17 Smoke aerosol response The monitor shall not spuriously alarm when the monitor is exposed to aerosols of 0 5 μm size, in concentrations of one microgram per liter.

3 4 18 Dimensions. The monitor shall not exceed 0 7 cubic feet in volume (excluding auxiliary equipment) The stowage case shall not exceed 1 3 cubic feet in volume and shall be capable of being passed through a 63 50-centimeter (25-in) diameter hatch

3 4 19 Weight The weight of the monitor shall not exceed 11 8 kilograms (kg) (26 pounds (lbs)) (including power cord and battery) The stowage case, including monitor and auxiliaries (see 3 3 3), shall not exceed 19 3 kg (43 lbs)

3 4 20 Ambient pressure The monitor shall be constructed so that ambient pressure changes (such as those due to snorkeling) within plus or minus 2 lbs of atmosphere (14 7 lb/in^2) absolute will not cause the alarm to sound

3 4 21 Moisture protection Components sensitive to moisture (resistors, insulators, switches, circuit boards, and so forth) may be sealed or coated to provide moisture protection In addition, housings containing these elements shall be provided with readily replaceable desiccant capsules to maintain dry atmospheric conditions Desiccant capsules shall contain a color-change indicator The monitor shall not require warm-up operation in order to dry up moisture on critical components

3 4 22 Enclosure The monitor shall be watertight as specified in MIL-STD-108 During test for watertightness, air inlet and air outlet shall be sealed off (see 4 6)

3 4 23 Closed-loop-operation The monitor shall be capable of operation in a closed-loop-configuration (that is, air outlet connected to air inlet to facilitate calibration, special monitoring or testing requirements, venting, and other special conditions requiring this type operation)

3 4.24 Temperature The monitor shall be designed to conform to the temperature requirements for range 4 of MIL-STD-2036 Thermal design shall be in accordance with MIL-STD-2036 The internal rise in temperature shall not exceed 10 degrees Celsius (°C) above ambient after 8 hours of continuous operation.

3 4 25 Temperature burn-in Monitors shall be subjected to a temperature burn-in with the environmental temperature equal to or greater than 90°C (see 4 6 8).

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3 4 26 Humidity. The monitor shall be constructed to conform to the humidity requirements specified in MIL-STD-2036.

3 4 27 Shock and vibration The monitor shall withstand the type A, class I, grade A shock test of MIL-S-901 for lightweight equipment and the type I vibration test of MIL-STD-167-1.

3.4.28 Controls, indicators, and panel layouts The controls and indicators design and the arrangement of controls and indicators on operator and maintainer panels shall conform to the criteria of MIL-STD-1472 and shall be as specified in the "Controls, indicators and panel layout" paragraph of MIL-STD-2036, with the following exceptions:

(a) Color coding of lamps and indicators shall be in accordance with MIL-STD-1472.
(b) Color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415

Abbreviations in accordance with MIL-STD-12 shall be used for monitor labeling, where applicable

3 4 28 1 Calibration switch The monitor shall have the capability of being switched to a calibrated microcuries per liter range for operation with the existing converters (IC-T2/FC and IC-T2/FC(A))

3 4 29 Electromagnetic interference (EMI) control EMI control shall conform to requirement 61 of MIL-STD-454 The monitor shall conform to the requirements for class A4 equipment as specified in MIL-STD-461, except as otherwise specified in 4 6 6 MIL-HDBK-241 may be used for guidance in design of power supplies.

3 4.30 Safety Safety requirements shall be in accordance with MIL-STD-2036 except, for the purposes of this specification, the interlock indicator light shall not be required (see 6.3).

3 5 Quantitative reliability requirements The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 3000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1000 hours.

3 6 Maintainability The monitor shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 30 minutes for repair (corresponding to the intermediate level of maintenance) that is accomplished by replacement of parts (electronic, electrical electromechanical, or mechanical parts) The corrective maintenance time shall include localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks

3.7 Workmanship. Workmanship shall conform to the workmanship and soldering requirements of MIL-STD-2036.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except

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as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

4.1 1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3) The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

4 1 2 Quality assurance terms and definitions Quality assurance terms used in this specification shall be as defined in MIL-STD-109

4 2 Classification of inspections The inspection requirements specified herein are classified as follows

(a) First article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4 3 Test conditions Unless otherwise specified, tests shall be performed at factory ambient temperature, humidity, and pressure Input power shall be 115 V, 60-Hz, single-phase, type I electric power in accordance with MIL-STD-1399, section 300

4 4 First article inspection Unless otherwise specified (see 6.2), four monitors shall be required for first article inspection First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examinations and tests specified in table I (see 6 3)

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TABLE I. First article inspection.

Examination or test	Requirement	Test method
Surface examination		
Weight	3 4 19	4 6
Dimensions	3 4.18	4 6
Parts and materials	3 2 2	4 6
Finish	3 2.2	4 6
Marking	3.2	4.6
Workmanship	3 7	4 9
Safety	3 4.30	4 6
Performance test	3 4 4 1, 3 4 4.2, 3 4 5 1 through 3 4 17, 3 4 23	4 6
Temperature test	3 4 24	4 6 1 1
Humidity test	3 4 26	4 6 1.2
Dielectric withstanding voltage test	3 2	4 6
Enclosure test	3 4 22	4.6
Shock test	3 4 27	4.6.2
Vibration test	3 4 27	4 6.3
Thermal design test	3 2	4.6 5
Power	3 4 4	4 6
Ambient pressure test	3 4 20	4.6 4
EMI test	3 4.29	4 6 6
Dripproof test	3 2	4.6
Salt spray test	3.2	4 6.7

4 5 Quality conformance inspection Quality conformance inspection shall be as specified in 4 5 1 through 4 5.3 1

4 5 1 Production inspection Production inspection shall be made on every monitor offered for delivery Production inspection shall comprise examinations and tests that will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials Production inspection shall include the examinations and tests shown in table II

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TABLE II. Production inspection.

Examination or test	Requirement	Test method
Surface examination		
Weight	3.4.19	4.6
Dimensions	3.4.18	4 6
Parts and materials	3.2.2	4.6
Finish	3.2.2	4 6
Marking	3.2	4 6
Workmanship	3.7	4.9
Performance test	3 4 4.1, 3 4.4.2, 3 4.5.1 through 3 4.17, 3 4.23	4 6
Temperature burn-in	3 4 25	4 6 8

4 5 2 Production control inspection Production control inspection shall be as specified in table III with acceptance based on requirements of table IV Production control inspection shall be performed on monitors that have passed production inspection The equipment shall conform to the requirements of production control inspection prior to release for shipment (see 6 6)

TABLE III Production control inspection

Inspection	Requirement	Test method	
Supply line voltage and frequency	3.2	4 6	
Thermal design test	3 2	4 6, 4 6 5	
Enclosure test	3 4 22	4.6	
Power	3 4 4	4 6	

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TABLE IV Sampling for production control inspection.

Lot size	Sample size	Accept	Reject 1/2/3/
Major defects			
2 - 50	2	0	1
51 - 500	8	1	2
501 - 3200	13	2	3
3201 - 35000	20	3	4
Minor defects			
2 - 150	5	1	2
151 - 500	8	2	3
501 - 3200	13	3	4
3201 - 35000	20	5	6

1/ Defective items must be replaced with acceptable items prior to lot acceptance
2/ Inspect sample size until reject criteria is reached
3/ Rejected lots may be screened and resubmitted for inspection and retest

4.5 3 Environmental inspection Environmental inspection shall be accomplished on a sampling basis as specified in 4.5 3 1 Environmental inspection shall be conducted to prove the durability of the monitor to withstand the effects of extreme environment, service conditions, and hard usage Environmental inspection shall be performed on monitors that have passed the production inspection (see 4 5 1) and shall include the examinations and tests shown in table V

TABLE V Environmental inspection

Examination or test	Requirement	Test method
Temperature test	3.4.24	4 6.1.1
Humidity test	3 4.26	4.6.1.2
Shock test	3 4.27	4 6.2
Vibration test	3 4 27	4 6 3

4 5.3 1 Environmental inspection samples Samples for the environmental inspection shall be

(a) One sample selected at random from the first 25 production units fabricated.
(b) One sample selected at random from each successive 100 units, or fraction thereof, produced

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4.5.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control and environmental inspections may be accepted on the contract provided the sample units are resubjected to, and pass, production inspection (see 4.5 1) after repair of all visible damage.

4 6 Test methods. The examinations and tests specified in table I shall be conducted in accordance with the applicable test methods of MIL-STD-2036, except as specified in 4 6 1 through 4.6 8

4.6.1 Temperature and humidity Temperature and humidity tests shall be conducted as specified in 4.6.1.1 and 4 6.1.2

4 6 1 1 Temperature tests. The monitor shall be capable of satisfactory operation over the temperature range specified in 3 4.24 The monitor shall indicate within the accuracy requirements of 3 4 9 over the entire cycle The equipment alarm (see 3 4 6) shall be tested at approximately 15 degree increments to verify functional performance over the temperature range

4 6 1 2 Humidity The humidity test shall be conducted in accordance with method 507 2, procedure III of MIL-STD-810 to determine conformance to 3 4 26 The monitor shall be in closed-loop-configuration during the test

4 6 2 Shock test The monitor shall be shock tested to determine compliance with the requirements specified in 3 4 27 The monitor shall be stowed in its stowage case with power off and in normal position for each blow Following each blow, the monitor shall be checked for accuracy, alarm, and circuit test

4 6 3 Vibration test The monitor, secured in the stowage case with the power off, shall comply with the vibration test of MIL-STD-2036 While under test, the monitor shall be in the normal stowed position The variable frequency test shall be performed with the monitor out of the stowage case The monitor shall be placed on the vibration table in normal operating position The monitor shall be allowed to move freely on the table, with only a minimum of restriction to prevent a tipover or rotation, and shall be tested to determine compliance with the requirements specified in 3 4 27 The vibration time at each integral frequency shall be 1 minute The table amplitudes shall be in accordance with the amplitudes of vibration specified for type I test in MIL-STD-167-1 All three axes shall be tested

4 6 4 Ambient pressure The monitor shall be subjected to five cycles of air pressure variation between the limits of 12 7 and 16 7 lb/in^2 absolute within a 30-second period The alarm shall not sound during this test

4 6 5 Thermal test. Equipment shall be operated continuously with full power in an ambient temperature of approximately 23 ± 5°C for at least 8 hours. During this period, the temperature inside the housing shall not exceed the ambient temperature by more than 10°C

4 6 6 EMI tests. EMI emission and susceptibility and self-compatibility tests for first article inspection shall be performed in accordance with MIL-STD-462 Inspections shall include tests CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 of MIL-STD-462, with the following exceptions

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(a) Test CS06 shall be performed at 400 V peak 5 microseconds wide.
(b) Test RS03 shall be performed at 50 V/m from 14 kilohertz to 30 megahertz (MHz) and 10 V/m from 30 MHz to 10 gigahertz

4 6.7 Salt fog test The monitor shall be subjected to the salt fog test of method 509.2 of MIL-STD-810.

4.6.8 Temperature burn-in The equipment shall be subjected to a temperature burn-in with the environmental temperature equal to or greater than 50°C with the equipment operating in the worst-case mode of a cumulative operating time of 48 hours (see 6 3). Failures occurring during the last 24 hours shall be repaired and burn-in resumed with a minimum of 48 hours of equipment conditioning without further failure Equipment monitoring during burn-in shall include, as a minimum, the functional checks to ensure proper operating under the worst-case operating mode No adjustment of the equipment (for example, manual compensation for drift in voltage controlled oscillators), other than normal operating mode changes, shall be allowed during the burn-in

4 7 Reliability testing The reliability of the monitor shall be demonstrated in accordance with the requirements of MIL-STD-781 and MIL-HDBK-781 (see 6 3)

4 7 1 Failure Failures (see 6 4.1) shall all be considered relevant unless determined by the contracting activity to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service

4 7 2 Performance parameters for reliability testing The parameters to be monitored shall include the following

(a) Background (see 3 4 7)
(b) Gamma compensation (see 3.4.8)
(c) Accuracy (see 3 4 9)
(d) Response time (see 3 4.10).
(e) Warm-up time (see 3 4.11)
(f) Power failure (see 3 4 4.2).
(g) Alarm (see 3 4.6).
(h) Remote air monitoring (see 3 4 14)
(i) Circuit check (see 3 4 12)
(j) Decontamination (see 3 4.15)
(k) Smoke aerosol (see 3 4 17)
(l) Closed-loop-operation (see 3 4.23)

4 7.3 Reliability verification tests Reliability verification testing shall be conducted only on the first production lot. Testing shall be performed using test plan IV of MIL-HDBK-781 at the following test level

Temperature	50 to 55°C (122 to 131 degrees Fahrenheit (°F))
Temperature cycling	None

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Vibration	The vibration shall be 1.0G ± 10 percent peak acceleration value at any nonresonant frequency between 20 and 60 cycles per second measured at the mounting points of the equipment The duration of vibration shall be at least 10 minutes during each hour of equipment operating time Vibration shall be normal to the plane of the majority of the cards. This vibration requirement may by met by mounting the equipment solidly, (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plate and driver by a suitable motor
Equipment on-off cycling	Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test
Input voltage	115 V, 60-Hz, single-phase, type I electric power in accordance with MIL-STD-1399, section 300
Input voltage cycling	The input voltage shall be maintained at 110 percent nominal for one-third of the equipment "on" cycle, at the nominal value for the second one-third of the equipment "on" cycle, and at 90 percent for the final one-third of the equipment "on" cycle This cycling procedure is to be repeated continuously throughout the reliability test

Verification testing shall be performed on four production units. Reliability testing shall be successfully completed prior to equipment shipment.

4 7 4 Reliability sampling tests Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment Sampling testing shall be performed using test plan VI of MIL-HDBK-781 with the same test level as described in 4 7 3 MIL-STD-781 and MIL-HDBK-781 shall be used to determine the number of units to be selected for reliability sampling tests Reliability testing for each lot shall be successfully completed prior to equipment shipment

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4.8 Maintainability demonstration. Demonstration of corrective and preventive maintenance shall be performed prior to the delivery of production equipment (see 6.3).

4 8 1 Corrective maintenance The demonstration of corrective maintenance shall be performed to determine conformance with the requirements of 3.6 The demonstration shall consist of 20 corrective maintenance actions involving faults selected at random from 40 candidate faults. The candidate faults shall be determined in accordance with appendix A of MIL-STD-471. Faults shall be created by the introduction of faulty parts or by simulation thereof.

4 8.2 Preventive maintenance Preventive maintenance tasks shall be performed to identify any problems and verify the adequacy of instructions contained in the technical manual (see 6 4.1).

4 8.3 Technical documentation Technical documentation to be used in the demonstration shall be limited to the technical manual (see 6 4 1).

4 8 4 Calculation of M_{ct} The demonstrated M_{ct} shall be calculated by averaging the times of the 20 corrective maintenance actions (see 6 6)

4 9 Workmanship The monitor, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements specified in 3 7 Each solder connection and its associated wiring or leads shall be visually examined

4 10 Inspection of packaging Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5 1 Preservation, packing, and marking. Unless otherwise specified herein, preservation, packing, and marking shall be in accordance with MIL-E-17555 (see 6 2)

6 NOTES

(This section contain information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The monitor covered by this specification is intended for use ashore and onboard ships

6 2 Acquisition requirements Acquisition documents must specify the following.

(a) Title, number, and date of this specification

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(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)
(c) When first article inspection is required (see 3 1)
(d) When the number of first article samples is other than as specified (see 4 4)
(e) When reinspected production control and environmental sample units may not be accepted (see 4 5 4)
(f) Level of preservation, packing, and marking (see 5.1).

6 3 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27 475-1 exempts the requirement for a DD Form 1423

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3 4 30	DI-R-5297	Quality program plan (QPP)	----
4 1 1	DI-SAFT-80102	Safety assessment report	----
4 4	DI-NDTI-80603	Test procedure	Subtitle CDRL "first article inspection procedure"
4 4	DI-NDTI-80809	Test/inspection reports	Subtitle CDRL "first article inspection report"
4 6 8	DI-RELI-80249	Environmental stress screening (ESS) report	----
4 7	DI-R-7079	Reliability program plan	----
4 7	DI-R-7085	Failure mode, effects and criticality analysis report	----
4 7	DI-RELI-80250	Reliability test plan	----
4.7	DI-RELI-80252	Reliability test reports	----
4.8	DI-MNTY-80822	Maintainability program plan	----
4 8	DI-MNTY-80827	Maintainability predictions report	----

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4.8	DI-MNTY-80828	Maintainability analysis report	----
4 8	DI-MNTY-80831	Maintainability/testability demonstration test plan	----
4 8	DI-MNTY-80832	Maintainability/testability demonstration test report	----

The above DID's were those cleared as of the date of this specification The current issue of DoD 5010 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6 4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3 1), and the number of items to be tested as specified in 4 4 The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6 4 1 Technical manuals The requirement for technical manuals should be considered when this specification is applied on a contract If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract

6.5 Definition The definition of failure used in this specification is given in 6 5.1

6 5 1 Failure Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements of this specification

6 6 Maintainability acceptance or rejection.
The demonstrated M_{ct} should be compared to the specified M_{ct} to determine acceptance or rejection If a reject decision is reached, the contracting activity should be immediately notified The contractor should, at no additional cost to the Government, correct all deficiencies and repeat the demonstration until an accept decision is reached

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6 7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6 8 Subject term (key word) listing

Detection chamber
Ion trap
Tritium converter

6.9 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity
Navy - SH
(Project 6665-N595)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-T-23752E(SH)	92/05/04

3 DOCUMENT TITLE: TRITIUM AIR MONITOR, PORTABLE OR FIXED

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
Technical Point of Contact (TOPC): Mr. Ron Rhodes (SEA 04RL) LEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	(1) Commercial TOPC: 703-602-2753	(2) AUTOVON 8-332-2753
c ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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